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ABSTRACT

In stock market forecasting achieving good prediction accuracy is always been a highly challenging task for researchers and financial analyst. Forecasting stock market needs to deal with the most volatile, non-parametric and non-linear data sets. Also there are various factors that may affect the growth of stock market. So in order to make a good stock market forecasting system we need to use all the parameters that may affect the market volatility. Support Vectors Machine (SVM) have been found to be one of most efficient machine learning algorithm in modeling stock market prices and movements. Researchers are using these classification algorithms for so many years and have got a good predictive accuracy. Here in our research we have used SVM algorithm to making prediction for CNX NIFTY index value. In our experiment we have compared prediction accuracy for various Kernel Types of SVM.

Keywords

SVM algorithm, stock market, index value, kernel types

1. INTRODUCTION

Stock market forecasting is in itself a very complex problem but it's really worthy to work on this problem because it have enough potential to make a big impact on policy making strategies of a country. Researchers have used various methods for making the predictions of market volatility but still getting a good accuracy is not a cupcake.

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It involves working on volatile, non-parametric, dynamic, chaotic and nonlinear data. To make predictions for the volatility of stock market we need to consider all the factors that may affect the performance. Researchers try to build models that can make prediction about the volatility of market using smaller set of data. Because here for this problem, there are a lot of different factors that can make impact on the market but using all this data will make the problem more tenuous.

Using Artificial Intelligence (AI) techniques for making predictions has made the task very easier for researchers, as they provide a very high accuracy compared to statistical and other techniques. In AI techniques, Artificial Neural Network (ANN) and Support Vector Machine (SVM) are two most powerful methods that can be used for solving the stock market forecasting problem. Both of these methods have their own strengths. However researchers have been proved that although ANN can be one of the most efficient AI algorithm for time series prediction but for stock market prediction it is not that much suitable because of the fact that data for stock market trades is non stationary and contains some noise in it. Using ANN often shows unpredictable and inconsistent performance on noisy data [7, 10, 11, and 14].

In our previous work while surveying about intelligent system approaches for stock market forecasting we have found that support vector machine provides a much better accuracy as compared to ANN or other AI models on noisy data [3]. Support vector machine works on Structural risk minimization (SRM) to develop binary classification which helps in regulating generalization and finding an optimal hyperplane. That's why it provides higher accuracy as compare to other models like ANN which works on Empirical risk minimization (ERM) [3]. Also it needs more computational resources as compared to others, it is still a good classifier. Support vector machine is also useful for some other classification and prediction based problems. For example, it is also useful for genomic predictions, protein classification, bankruptcy predictions, etc. [2].

2. Input Data Sets

For every classification model data set is first and most important thing that is required. Researchers have been used various kind of data set with in different range of time durations. In the process of designing our classification model we have used 1046 data samples of CNX Nifty index from 23 May, 2011 to 30 December, 2015. All the data is acquired from yahoo finance website. Our data samples consist of values of opening, high, low and closing price of each day. We have used these basic input variables for extracting technical features for our classifier. Some researchers also use volume as data input variable. Here in our classification model we have been used 5 fold cross validation model [18], so for every training data set we used 80 % of our whole data for training and rest as validation data set.

3. Data Preprocessing

Raw data that we got from our data sources contains inconsistency and noise. The quality of data that we used for modeling affects accuracy of the prediction model. For making the data ready for modeling process first it needs to be preprocessed. For preprocessing of data we normalized each input data instance into a linearly scaled specified range [12].

To converting data into scaled range researchers use to convert data into fix specify ranges like [-1, 1], [0, 1], [-2, 2] etc. Here for our modeling, we have scaled our data in a

range of [0, 1] this will reduce the complexity involved in data due to inconsistent range of data.

4. Feature Extraction

Feature extraction is also a very important part of modeling a classifier. Extracting good features from raw data will definitely increase the accuracy of the classification model. Size of feature vectors makes a huge impact on the accuracy of the classifier. Researchers are using various kinds of features as input variables. For making prediction about long term trades researchers use fundamental indicators like macroeconomic data, interest rates, money supply, exports, imports etc.

While here we are modeling our classifier for short term predictions, so we used technical indicators [3]. Various fund manager and investors in stock market historically accepted and used established criteria for extracting technical indicators as a sign for market volatility [14].

We have extracted total 20 features from our raw data set that we used in our experiment. These are Open, Close, Low, High, SMA 12, SMA 26, SMA 100, EMA 12, EMA 26, EM100, MACD, Signal, Histogram, Change, Gain, Loss, Avg. Gain, Avg. Loss, RS, RSI. These technical indicators will show the overall movements and volatility of the stock market index.

Here in table 1 we have been provided a list of all the technical indicators that are widely used by the researchers for market prediction.

Table 1: Table of Technical Indicators

Name of indicator	Description	Formula
Simple 12 Day Moving Average(SMA12)	It is a simple mean for 12 days stock index closing prices.	$\frac{C_{12} + C_{11} + C_{10} \dots \dots + C_1}{12}$
Simple 26 Day Moving Average(SMA26)	It is a simple mean for 26 days stock index closing prices.	$\frac{C_{26} + C_{25} + C_{24} \dots \dots + C_1}{26}$
Simple 100 Day Moving Average(SMA100)	It is a simple mean for 100 days stock index closing prices.	$\frac{C_{100} + C_{99} + C_{98} \dots \dots + C_1}{100}$
Exponential 12 day Moving Average(EMA)	It is similar to SMA except the fact that here we give more weightage to recent day prices	$C_{12} \times 2/(12 + 1) + EMA_{11} \times (1 - 2/12 + 1)$
Exponential 26 day Moving Average(EMA)	It is similar to SMA except the fact that here we give more weightage to recent day prices	$C_{26} \times 2/(26 + 1) + EMA_{25} \times (1 - 2/26 + 1)$
Exponential 100 day Moving Average(EMA)	It is similar to SMA except the fact that here we give more weightage to recent day prices	$C_{100} \times 2/(100 + 1) + EMA_{99} \times (1 - 2/100 + 1)$
Moving Average Convergence Divergence(MACD)	It is a momentum indicator. It shows relationship between two moving averages.	$MACD(n)_{t-1} + 2/(n + 1) \times (DIFF_t - MACD(n)_{t-1})$
Signal Line	It is nothing but 9 th day EMA of MACD	$MACD(EMA_9)$

Histogram	It measures the distance between MACD and its signal line	MACD – Signal Line
Relative Strength	Average of x days' up closes / Average of x days' down closes.	<i>Average Gain/Average Loss</i>
Relative Strength Index	It is a momentum oscillator that measures the speed and change of price movements. It oscillate between 0 and 100.	$100 - \frac{100}{1 - RS}$

5. EXPERIMENT SETUP

Here in this section we have been discussed about the experimental setup and classification model in detail. We have explained about the SVM classifier that we are going to use, data set.

5.1 SVM Classifier

Support vector machine is introduced by Boser, Guyon and Vapnik in 1992[1]. It is a kind of state-of-the-art- classification method which can be applied to almost all kind of classification problems. SVM is categorized into support vector classification (SVC) for classification related problems and support vector regression (SVR) for regression problems. In support vector machine we try to find a hyperplane that separate data into different classes. It classifies data by detecting an optimal hyperplane that separates data points of one class from data points of the other class. Here by an optimal hyperplane we mean that it has largest margin between the two classes. Margin means the maximum width parallel to hyperplane in which there is no data point. The data points closest to this separating hyperplane are known as support vectors.

As an example for a training set with input vectors $x_i \in R^d$ ($i=1, 2, N$) and their matching labels $y_i \in [+1, -1]$ ($i=1, 2, \dots, N$). Support vector machine learns to classify the various objects into two separate classes. Here the two classes are +1 and -1. Here SVM classifier maps the input vectors $x_i \in R^d$ into a high dimensional feature space $\phi(x_i) \in H$ and fabricate an Optimal Separating Hyperplane (OSH), which maximizes the distance between hyperplane and the nearest data points from each of classes in the space H, known as margin [7]. A linear SVM model and its hyperplane is shown in the figure 1 for giving overall idea hyperplane and margin in support vector machine [6]. One of the most important quality of the SVM is it's concept of kernel functions, by which we can solve problems for data with very high number of feature and can achieve highly optimized hyperplane [19]. Linear kernel is used for linear data but for non-

linear data some other kernels are introduced. Two most widely used kernels are Polynomial kernel and Radial Basis Kernel. [4, 5].

5.2 Classification Model

Predicting stock market index is in itself very tough job because of the fact that here our data is non-stationary in nature. The stock market trades at a particular time can be cycles, random walks, trends or combination of all the three. A good prediction system should be able enough to predict both bullish and bearish trades for the stock market [12].

For stock market prediction system SVM has always been the most preferred method. Various researchers have been used SVM for stock market prediction system [12, 13, 14, 15, 16, 17].

Stock market forecasting is a two class problem. The direction for daily changes at stock market is categorized in two classes. When value of CNX Nifty Index at time x is higher than that at time x-1, it will be came in category of “+1”, while if value of CNX Nifty Index at time x is lower than that at time x+1, it will be classified in “-1” category [7]. Now the real raw data is normalized into range of [0, 1]. By scaling of data using normalization we get surety that smaller value attributes will not be overwhelmed by larger value attributes, also it will reduce error [10, 11].

Here in our experiment we have studied and applied all four kernels. We have been build classification models using each and every kernel. For the implementation of machine learning algorithms we will use Orange data mining tool [9]. This tool provides us all the needed kernel setups with their parameters. For validation of classification model here we have been used K-fold validation method. Here we have chosen 5 as value of K.

Table 2: Types of Kernel in SVM

Kernel Type	Expression
Linear Kernel	$K(x_i, x_j) = x_i \cdot x_j$
Polynomial Kernel	$K(x_i, x_j) = (x_i \cdot x_j + 1)^d$

Hyperbolic Tangent Kernel	$K(x_i, x_j) = \tanh(c_1(x_i \cdot x_j) + c_2)$
Radial Basis Kernel	$K(x_i, x_j) = \exp(- x_j - x_i /2p^2)$

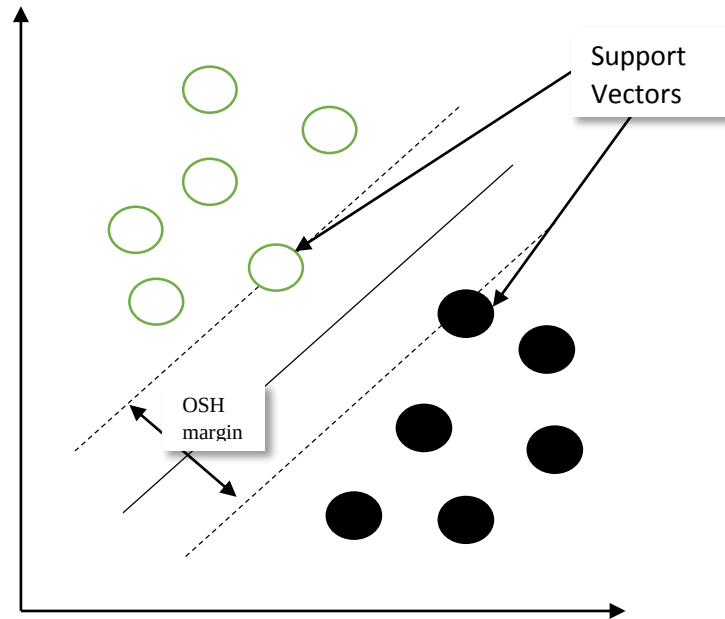


Figure 1: SVM with linear Kernel

In linear kernel SVM we have a big advantage over other models that we needn't to choose any parameter to tune the classifier except the constant C. In case of nonlinear SVM we have one more additional parameter known as Kernel parameter to tune for building the classifier [20]. From our study we came to know that polynomial classifier takes longer time in training and provides less accuracy as compare to Radial basis Kernel. From our experiments we came to know that linear SVM gives highest accuracy, other than linear kernel radial basis kernel also provides a good accuracy for prediction of market. While polynomial kernel provides an average accuracy and tangent kernel gives a very low accuracy of only 62%. It proves that linear kernel and Gaussian based radial basis kernel are most suitable methods for stock market prediction system.

Table 3: Accuracy for various classifiers

Types of SVM Kernel	Accuracy
Linear	88.34 %
Polynomial	72.54 %
Radial Basis	82.1 %

Hyperbolic Tangent Kernel	62.11 %
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Conclusion

From our research experiments we have been reached at few conclusions. First of all that SVM is most suitable method for stock market forecasting problem. Also in SVM Linear kernel provides a very good accuracy because of that fact that the number of features and data points is not so high. Due to the same reason Radial basis kernel also provides a good accuracy. For future research it will be a really good idea to research on connections between the number of features, data points and accuracy of the classifier. Also using some fundamental indicators will definitely increase the accuracy of the classification model. Furthermore one can study in deep effect of adjusting various parameters in SVM kernels.

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